DIAGONAL SHEAF IN THE THICK HULL OF A BOX PRODUCT

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Resolution property. perfect-lemma-regular-resolution-property (cross-chapter, named). Smooth implies the resolution property.
- Diagonal resolution. lemma-diagonal-resolution L26494 (carded this batch). Provides the box-product resolution.
- Regular split. lemma-split-complex-regular L26491 (carded this batch). Splits the two-step complex, isolating O_Delta.

Source reference. Original reference 0FX9: focused-seeds/geometry/S9-eqv-diagonal-trick/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0FX7: focused-seeds/geometry/S9-eqv-diagonal-trick/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09SI: focused-seeds/geometry/S9-eqv-diagonal-trick/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FZ2: sources/stacks-equiv/pinned-original-source.tex, source lines 1836–1836. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FZ4: sources/stacks-equiv/pinned-original-source.tex, source lines 1920–1920. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0F8A: focused-seeds/geometry/S9-eqv-diagonal-trick/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 056S: focused-seeds/geometry/S9-eqv-diagonal-trick/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let X be a quasi-compact separated smooth scheme over k. There exist finite locally free $\mathcal{O}_X$-modules $\mathcal{E}$ and $\mathcal{G}$ such that*

$$\mathcal{O}_\Delta \in \langle \mathcal{E} \boxtimes \mathcal{G} \rangle$$

in $D(\mathcal{O}_{X\times X})$ where the notation is as in Derived Categories, Section 09SI.

Proof. Recall that X is regular by Varieties, Lemma 056S. Hence X has the resolution property by Derived Categories of Schemes, Lemma 0F8A. Hence we may choose a resolution as in Lemma 0FZ2. Say $\dim(X) = d$. Since $X \times X$ is smooth over k it is regular. Hence $X \times X$ is a regular Noetherian scheme with $\dim(X \times X) = 2d$. The object

$$K = (\mathcal{E}_{2d} \boxtimes \mathcal{G}_{2d} \to \ldots \to \mathcal{E}_0 \boxtimes \mathcal{G}_0)$$

of $D_{perf}(\mathcal{O}_{X\times X})$ has cohomology sheaves $\mathcal{O}_\Delta$ in degree 0 and $\mathrm{Ker}(\mathcal{E}_{2d} \boxtimes \mathcal{G}_{2d} \to \mathcal{E}_{2d-1} \boxtimes \mathcal{G}_{2d-1})$ in degree $-2d$ and zero in all other degrees. Hence by Lemma 0FZ4 we see that $\mathcal{O}_\Delta$ is a summand of K in $D_{perf}(\mathcal{O}_{X\times X})$. Clearly, the object K is in

$$\Big\langle \bigoplus\nolimits_{i=0,\ldots,2d} \mathcal{E}_i \boxtimes \mathcal{G}_i \Big\rangle \subset \Big\langle \Big(\bigoplus\nolimits_{i=0,\ldots,2d} \mathcal{E}_i\Big) \boxtimes \Big(\bigoplus\nolimits_{i=0,\ldots,2d} \mathcal{G}_i\Big) \Big\rangle$$

which finishes the proof. (The reader may consult Derived Categories, Lemmas 0FX9 and 0FX7 to see that our object is contained in this category.) □